MACHINE LEARNING / NLP PROJECT

Indian Food Recognition and Recipe Generation

A pretrained vision model with regional knowledge and text generation

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Working prototype; text-output quality and model performance remain limited.



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Abstract

This project presents a web-based prototype that accepts an image of an Indian food item, predicts a label using a pretrained image-classification model, retrieves regional information and a reference recipe from a curated knowledge base, and offers an optional text-generation step using FLAN-T5-small. The application is implemented in Python with Streamlit and uses Hugging Face Transformers for inference. In the observed demonstration, an idli image was classified as Idli and the interface displayed South India as its likely regional association, together with a short description and typical preparation method. The text-generation component was integrated, but its output was not consistently satisfactory during testing; it is therefore treated as experimental rather than a validated recipe-writing system. The prototype demonstrates an end-to-end concept, while fixed classifier classes, curated regional facts, limited testing, and CPU inference are important constraints.

Keywords

Indian food recognition; image classification; Streamlit; Hugging Face; FLAN-T5; recipe generation.

1. Introduction

Food-image recognition can help users discover the name and preparation of a dish from a photograph. Indian cuisine is diverse, with regional names, ingredients, and preparation styles. This project demonstrates a small pipeline combining image classification, structured food knowledge, and natural-language generation.

2. Problem Statement

A user may have a food photograph but not know the dish name, common regional association, or typical preparation. The project explores a simple application that predicts a supported label and presents contextual information. A pretrained classifier recognises only its learned classes and cannot reliably identify every Indian dish.

3. Objectives

- Accept JPG/PNG food images in a browser interface.
- Predict a food label using a pretrained image classifier.
- Display regional association, description, and typical recipe from FOOD_INFO.
- Integrate FLAN-T5-small as an experimental text-generation component.
- Document implementation, observed results, limitations, and future work.

4. Scope and Assumptions

Included in the prototype	Not claimed by this prototype

Image upload and pretrained classification	A custom-trained or fine-tuned model
Curated regional facts and recipe text	Region inferred directly from pixels
Optional FLAN-T5-small generation	Consistently reliable, validated recipes
Local Streamlit interface	Production deployment
Limited supported labels	Recognition of all Indian foods

Regional associations are common or likely associations, not exclusive origins. Recipes vary by region and household; entries are illustrative.

5. System Architecture

Stage	Component	Output
1	User uploads JPG/PNG	Image file
2	Subhash5/indian-food-classifier	Candidate food labels and scores
3	FOOD_INFO dictionary	Region, description, reference recipe
4	google/flan-t5-small (optional)	Generated food-guide text
5	Streamlit UI	Results shown to user

The classifier and text model are separate components. The label is predicted from the image; regional information is retrieved from a curated dictionary; FLAN-T5 receives a prompt assembled from the known facts. These stages are not one jointly trained model.

6. Technologies

Component	Technology / model	Purpose
Interface	Streamlit	Upload controls and results UI
Image model	Subhash5/indian-food-classifier	Fixed-class food image classification
Text model	google/flan-t5-small	Experimental text generation
ML framework	Transformers + PyTorch	Model loading and inference
Image handling	Pillow	Read uploaded images
Runtime	Python virtual environment	Dependency isolation

7. Implementation

7.1 Image classification

The application passes an uploaded image to a pretrained image-classification pipeline and displays candidate labels and scores. A displayed score is not an estimate of overall model accuracy.

7.2 FOOD_INFO knowledge base

The FOOD_INFO dictionary maps food names to regional association, description, and typical recipe. This makes it easy to add information without retraining the vision model. Adding a

dictionary entry does not add a new visual class; for example, a Paani Puri recipe entry does not teach the classifier to recognise that dish.

7.3 FLAN-T5 text generation

The app loads google/flan-t5-small using AutoTokenizer and AutoModelForSeq2SeqLM, tokenises a prompt built from the food facts, generates tokens, and decodes the result. During development, the Transformers pipeline task name text2text-generation raised an error in the installed version. The implementation was changed to direct sequence-to-sequence model loading. A subsequent observed output was short and incomplete (for example, "Describe Idli, a popular Indian food."), so generation quality still requires validation.

7.4 Runtime

The app runs locally using Streamlit, typically at http://localhost:8501. Model files download from Hugging Face on first use and are cached for later runs. The observed setup used CPU inference, which can make initial loading and generation slower.

8. Demonstration and Observed Results

These are manual observations from development, not a formal benchmark.

Test	Observed result	Interpretation
Upload idli image	Image accepted	Upload path worked for this sample
Classification	Top label Idli; UI displayed 100.0%	One example only, not overall accuracy
Regional lookup	South India displayed	Retrieved from curated knowledge
Reference recipe	Description and typical recipe displayed	Lookup worked for the matched label
FLAN-T5	Model ran, but one output was short/incomplete	Integration exists; quality not yet validated
Unmatched labels	Some predictions may lack FOOD_INFO	Needs fallback or expanded dictionary

No held-out test set, confusion matrix, precision, recall, F1 score, or general accuracy measurement was completed. The 100.0% value was the displayed score for one image, not a claim that the system is 100% accurate.

9. Limitations

- Fixed class vocabulary: dishes outside the learned classes may be assigned an incorrect known label.
- Limited manual testing; performance on a representative dataset is unknown.
- Regional associations come from curated knowledge-base entries, not visual prediction.
- FLAN-T5-small may return short, repetitive, incomplete, or inaccurate text.
- Recipes are illustrative and should be checked for ingredients, cooking time, and food safety.
- Initial downloads and CPU inference may be slow.

- Predicted labels without a FOOD_INFO entry need a graceful fallback.

10. Future Enhancements

- Fine-tune or evaluate on a labelled, broader Indian-food dataset.
- Use a held-out test set, confusion matrix, and per-class precision, recall, and F1.
- Normalise labels and handle missing knowledge entries gracefully.
- Use curated recipe text as a fallback when generated text is empty or too short.
- Add trustworthy sources for regional and recipe facts.
- Add unit tests for classification lookup, unsupported labels, and empty output.
- Consider deployment only after testing, performance checks, and privacy review.

11. Conclusion

The project demonstrates an academic prototype connecting pretrained Indian-food image classification with curated contextual information and an experimental text-generation component. The local Streamlit interface and idli demonstration show that the core workflow can run. The fixed label set, curated regional information, limited evaluation, and inconsistent FLAN-T5 output define the current scope and provide clear directions for improvement.

12. References

- Hugging Face Model Hub, Subhash5/indian-food-classifier: https://huggingface.co/Subhash5/indian-food-classifier
- Hugging Face Model Hub, google/flan-t5-small: https://huggingface.co/google/flan-t5-small
- Transformers documentation: https://huggingface.co/docs/transformers/
- Streamlit documentation: https://docs.streamlit.io/
- PyTorch documentation: https://pytorch.org/docs/stable/

Appendix A. Run Instructions

```
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt streamlit
run app.py
```

Run these commands from the project folder. Adapt the activation command if your virtual environment is stored elsewhere. The first run may download model files.

Appendix B. Final checks

- Confirm student, roll number, course, faculty, and institution details before submission.
- Insert actual screenshots from the final app run; do not use mock images as evidence.

- Export the report to PDF and check page breaks, tables, and links.
- Upload app.py, requirements.txt, README.md, screenshots, report, and slides to the repository.
- Include only results actually observed during testing.